

Task 3 – Word Error Rate (WER)

Data & setup (manual):

- Language: Persian, conversational dialog.
- We annotated 60 lines (total 180 lines across the group) **by hand**.
- Columns added: **S**, **I**, **D** (and **N** = tokens in Ground Truth).
- WER formula: $WER = (S+I+D)/N$ (where N = tokens in Ground Truth).
- Punctuation not counted as tokens; ZWNJ/spelling differences counted as substitutions.

Results:

- **Whisper**: $S=314, I=23, D=45, N=845 \rightarrow WER = (314+23+45)/845 = 0.452 \approx 45.2\%$
- **Assembly**: $S=585, I=148, D=195, N=1274 \rightarrow WER = (585+148+195)/1274 = 0.728 \approx 72.8\%$

Comparison (paper, Table 13 – Persian):

- Persian WERs in the paper: **Small=55.8%**, **Medium=41.0%** (Large=36.1%, Large-v2=32.9%).
- Whisper WER (**45.2%**) sits **between Small and Medium** (better than Small, worse than Medium).

Conclusion:

- **Whisper** performs much better than **Assembly** on our data (45.2% vs 72.8%), so we will continue with Whisper in the pipeline.